

1. $A = 2x^2 - 3x - 3$, $B = 2x^2 - x - 4$, $C = 2x^2 - 2x + 1$ であるとき, 次の計算をせよ。

(1) $A + C$

答. _____

(2) $2A + C$

答. _____

(3) $A + B + C$

答. _____

(4) $A - B + C$

答. _____

(5) $3A - 3B - 2C$

答. _____

(6) $3A + 3B - 2C$

答. _____

(7) $3A + 2B - C$

答. _____

(8) $2A - B - C$

答. _____

2. 次の 1 次方程式を解け。

(1) $3x + 1 = -9x - 8$

答. _____

(2) $-3x - 6 = 5x + 8$

答. _____

(3) $-7 + 3(-9x - 2) = 8x$

答. _____

(4) $-2 + 2(-8x - 4) = x$

答. _____

(5) $2x - 8 + 5(-3x - 8) = 0$

答. _____

(6) $-6x - 1 = -\frac{9}{2}x - 8$

答. _____

(7) $7x + \frac{4}{3} = -\frac{8}{3}x - 1$

答. _____

(8) $\frac{1}{3}x + \frac{7}{3} = -2x + \frac{2}{3}$

答. _____

1. $A = 2x^2 - 3x - 3$, $B = 2x^2 - x - 4$, $C = 2x^2 - 2x + 1$ であるとき, 次の計算をせよ。

(1) $A + C$

答. $4x^2 - 5x - 2$

(2) $2A + C$

答. $6x^2 - 8x - 5$

(3) $A + B + C$

答. $6x^2 - 6x - 6$

(4) $A - B + C$

答. $2x^2 - 4x + 2$

(5) $3A - 3B - 2C$

答. $-4x^2 - 2x + 1$

(6) $3A + 3B - 2C$

答. $8x^2 - 8x - 23$

(7) $3A + 2B - C$

答. $8x^2 - 9x - 18$

(8) $2A - B - C$

答. $-3x - 3$

2. 次の 1 次方程式を解け。

(1) $3x + 1 = -9x - 8$

$12x = -9$

答. $x = -\frac{3}{4}$

(2) $-3x - 6 = 5x + 8$

$-8x = 14$

答. $x = -\frac{7}{4}$

(3) $-7 + 3(-9x - 2) = 8x$

$-7 - 27x - 6 = 8x$

$-35x = 13$

答. $x = -\frac{13}{35}$

(4) $-2 + 2(-8x - 4) = x$

$-2 - 16x - 8 = x$

$-17x = 10$

答. $x = -\frac{10}{17}$

(5) $2x - 8 + 5(-3x - 8) = 0$

$2x - 8 - 15x - 40 = 0$

$-13x = 48$

答. $x = -\frac{48}{13}$

(6) $-6x - 1 = -\frac{9}{2}x - 8$

両辺に 2 をかけて, $-12x - 2 = -9x - 16$

答. $x = \frac{14}{3}$

(7) $7x + \frac{4}{3} = -\frac{8}{3}x - 1$

両辺に 3 をかけて, $21x + 4 = -8x - 3$

答. $x = -\frac{7}{29}$

(8) $\frac{1}{3}x + \frac{7}{3} = -2x + \frac{2}{3}$

両辺に 3 をかけて, $x + 7 = -6x + 2$

答. $x = -\frac{5}{7}$